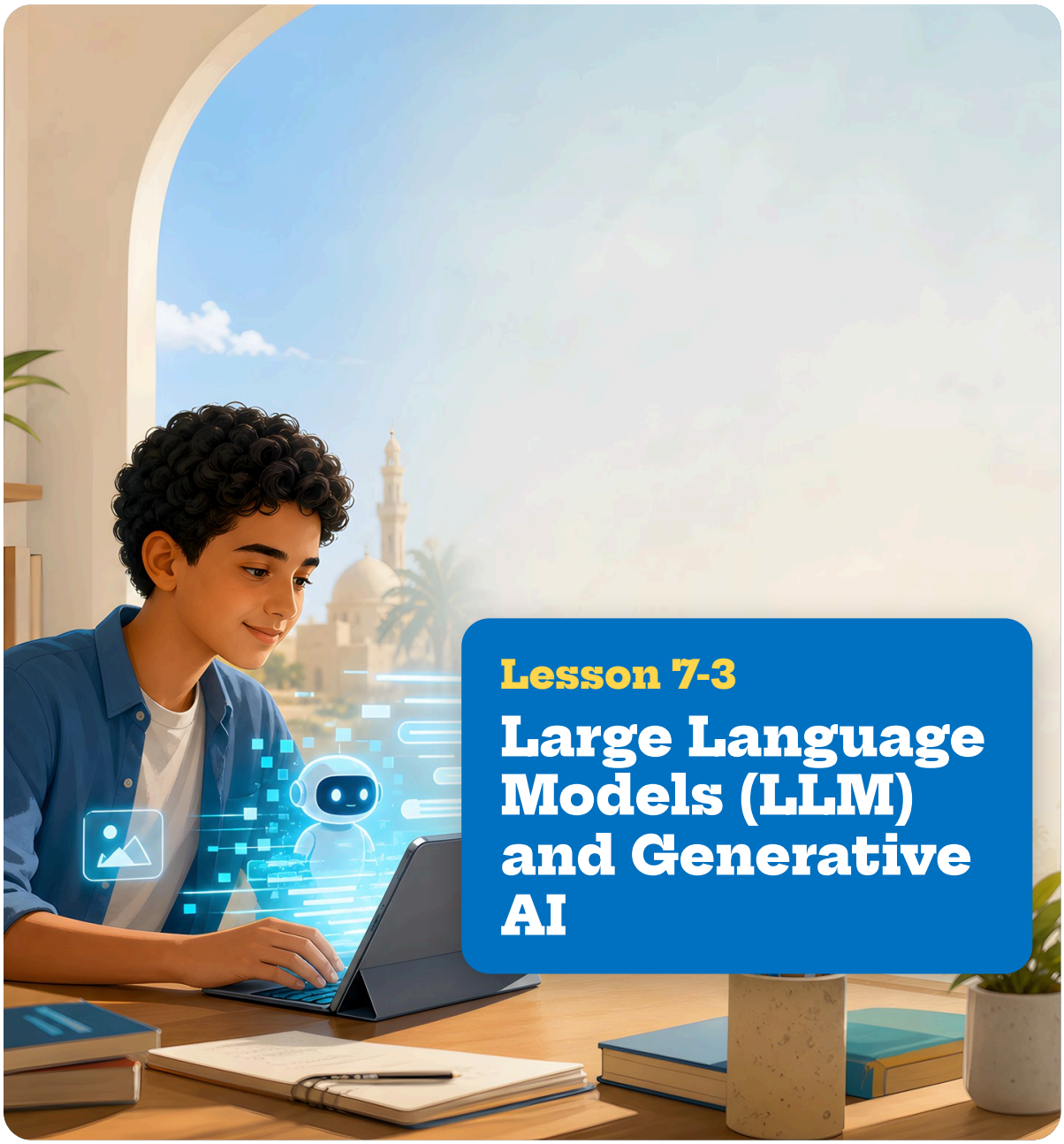


Learning Objectives

- Explain how a language model generates text by predicting the next word, and why this causes hallucination.
- Describe what makes a language model large, and explain the roles of RLHF tuning and multimodality.

When a chatbot writes a fluent paragraph, it is not looking up an answer — it is guessing the next word, then the next, using patterns learned from enormous amounts of text. That simple trick, scaled up to billions of parameters, gives large language models their power to translate, summarize, and converse. But because each word is chosen for being statistically plausible rather than certainly true, these models can state falsehoods convincingly — a failure called hallucination. This lesson shows how such models work, how human feedback (RLHF) makes them helpful, and how multimodality lets them handle images and audio too.

? Guiding question: How does a language model generate text, what makes a model "large," and why can it still get facts wrong?

Point!

1. Language Models and "Predicting the Next Word"

(1) **(Language model)**…a model created to learn from large amounts of text and predict which word is likely to come next in a sentence. Through training, it acquires patterns of word usage and phrasing according to context.

(2) Text generation is achieved by repeating the prediction of the next word.

Step	Input sentence	Word predicted next
1st	"Today's weather is"	"sunny"
2nd	"Today's weather is sunny"	"and"
3rd	"Today's weather is sunny and"	"warm"

In this way, a complete sentence is built up by connecting the results of predicting one word at a time.

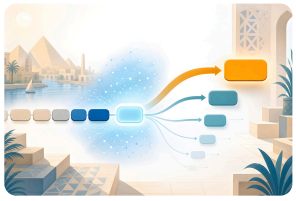
Key Fact:
A language model generates text by predicting the next word again and again — choosing what is statistically plausible, not guaranteed true (the cause of hallucination). Large language models scale this up with vast data and billions to trillions of parameters, are tuned by RLHF to be helpful, and, through multimodality, handle text, images, and audio together.

Key Concepts:

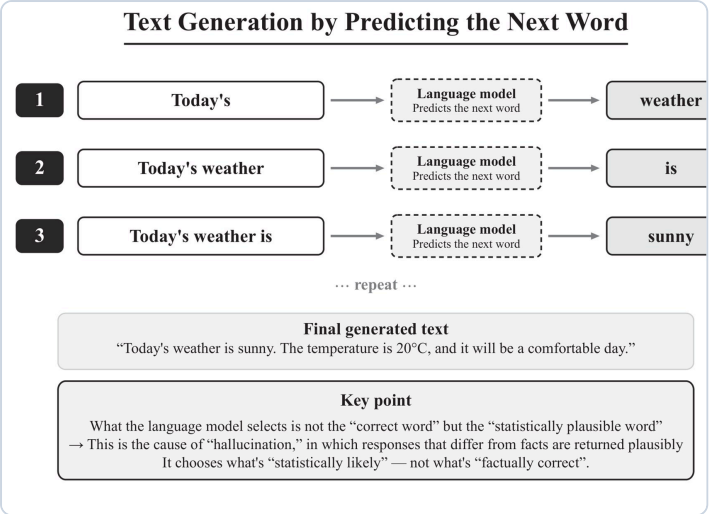
- language model, next-word prediction
- hallucination
- large language model (LLM), parameters
- RLHF
- multimodality

KEY TERMS

language model — predicts the likely next word.
hallucination — a plausible answer that is factually wrong.



A language model predicts the next word.



Text is generated by predicting the next word, one at a time

(3) For each candidate word, the language model calculates a "probability of coming next" and selects words based on that probability. These probabilities come from patterns the model learned during training, so it can also handle wordings that never appeared in the training data.

(4) What a language model selects is not the correct word, but a statistically plausible word. Because of this property, **(hallucination)**—the phenomenon of returning responses that differ from the facts but are stated in a plausible way—occurs.

2. Large Language Models (LLM)

(1) **(Large language model (LLM))**…a language model based on a very large neural network trained using vast amounts of text from the Internet.

(2) "Large" mainly refers to the following two kinds of size.

Aspect	Description
Amount of training data	The model is trained using vast amounts of text containing billions to trillions of words
Number of parameters	The numerical values corresponding to the weights of the neural network number in the billions to trillions

(3) Researchers have observed that when the number of parameters or the amount of training data exceeds a certain scale, qualitatively new abilities begin to appear. *e.g.,* handling complex reasoning; summarizing text; translating into foreign languages; generating program code

(4) Major services built on large language models include ChatGPT, Claude, and Gemini, which provide natural conversation and a wide range of text generation.

3. Tuning by RLHF

(1) Just by learning from vast amounts of text, large language models do not necessarily return responses that are desirable for humans. They may answer

KEY TERMS

large language model (LLM) — a huge network trained on vast text.

parameter — a weight; LLMs have billions to trillions.

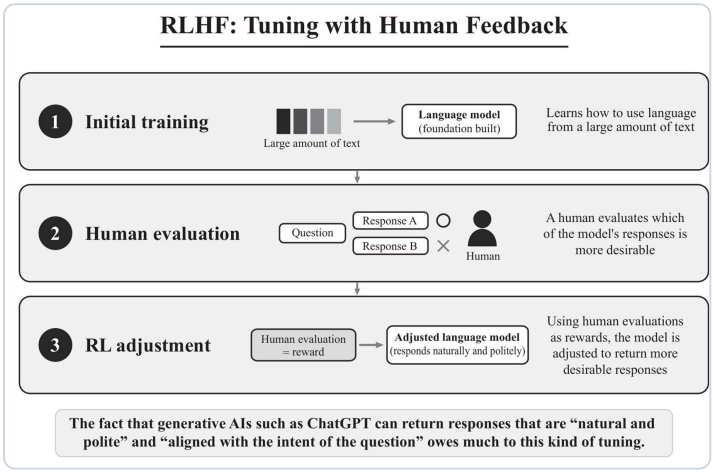
RLHF — tuning by human feedback as reward.

without grasping the intent of the question or return responses containing inappropriate content.

(2) A representative method for solving such problems is **(RLHF (Reinforcement Learning from Human Feedback))**. RLHF is a method that applies reinforcement learning to the tuning of language models.

(3) RLHF proceeds in the following flow.

Stage	Description
① **Initial training**	Build a language model that has learned word usage from large amounts of text
② **Human evaluation**	Humans give evaluations on the model's responses, such as "which is more desirable"
③ **Tuning by reinforcement learning**	Using human evaluations as rewards, tune the model to return more desirable responses



RLHF: human evaluations become rewards that tune the model

(4) Through RLHF, language models begin to return natural and polite responses, responses aligned with the intent of the question, and responses that avoid inappropriate content. The fact that generative AIs such as ChatGPT feel "easy to use" to users owes much to such tuning.

4. Multimodality

(1) **(Multimodality)**…the property of being able to handle multiple kinds of data with a single model, such as text, images, audio, and video. Each kind of data is called a **(modality)**.

(2) Early language models handled only text, but recent generative AIs can handle different kinds of data in combination. *e.g.,* looking at an image and describing its content in text; listening to audio and transcribing it into text; giving instructions in text and generating an image; summarizing the content of a video

(3) The background to multimodality being achieved is that neural networks have become able to handle different kinds of data by converting them into a

PAUSE & THINK

Left alone, a huge model may answer rudely or off-point. What step fixes that?



Human feedback ranks outputs to tune the model.

KEY TERMS

multimodality — one model handling many data kinds.
modality — one kind of data (text, image, audio…).

common numerical representation. Inside the network, text, images, and audio are all processed in the same way as sets of numerical values.

(4) Through multimodality, generative AI is evolving from a technology that handles only language into a technology that integrally handles the various kinds of information that humans deal with daily.



One model can output text, image, and sound.

PAUSE & THINK

If a model sounds confident and fluent, does that guarantee it is correct?

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A language model is a model that predicts which word is likely to come next in a sentence.

B A language model finishes processing after predicting just one word and does not generate text.

C In a large language model, when the number of parameters or the amount of training data exceeds a certain scale, qualitatively new abilities sometimes appear.

D Multimodality is the property of handling only text.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A language model based on a very large neural network trained using vast amounts of text from the Internet

b A method that uses human feedback as a reward to tune a language model to return more desirable responses

c The phenomenon in which a language model returns responses that differ from the facts in a plausible way

[Choices] **A** Hallucination **B** Large language model **C** RLHF

✓ Solution

(1)

A Correct definition of a language model. Therefore, ○.

B Text generation is achieved by repeating the prediction of the next word. Therefore, ×.

C Correct description of large language models. Therefore, ○.

D Multimodality is the property of being able to handle multiple kinds of data—text, images, audio, video, etc.—with a single model. Therefore, ×.

(2)

a: A huge language model trained on vast text is a large language model. **B**

b: Tuning by reinforcement learning using human feedback as a reward is RLHF. **C**

c: The phenomenon of returning fact-discrepant responses in a plausible way is hallucination. **A**

Try

1 Answer the following questions.

1. What is the term for the model that learns from large amounts of text and predicts which word is likely to come next in a sentence?
2. What is the term for a language model based on a very large neural network trained using vast amounts of text from the Internet?
3. What is the term for the method that uses human feedback as a reward to tune a language model to return more desirable responses?
4. What is the term for the property of being able to handle multiple kinds of data—text, images, audio, video, and so on—with a single model?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes how a language model works.

A It memorizes the input sentence as is and returns the same sentence.

B It generates text by repeatedly predicting the next word in the sentence.

C It selects and returns the closest match from prepared template sentences.

D It fully understands the meaning of the sentence and returns the single correct response.

(2) From the following options (A–D), choose the one that most appropriately describes the cause of hallucination.

A Because the language model always retrieves and returns sentences from the training data as is.

B Because the language model selects and returns words that are statistically plausible rather than correct.

C Because the language model avoids answering questions and returns unrelated text.

D Because the language model is built to return only fact-based responses.

EXAM-STYLE QUESTION

Explain why a language model can produce a fluent answer that is factually wrong (hallucination). **[6] Refer to:** choosing statistically plausible words.

PAUSE & THINK

"Plausible" or "correct" — which one does the model actually optimize for?

(3) From the following options (A–D), choose the one that most appropriately describes RLHF.

A A method of building a language model from scratch using a large amount of text.

B A method of tuning a language model to return more desirable responses, using human evaluations as rewards.

C A method that removes the neural network structure from a language model and simplifies the processing.

D A method that has a language model solve math problems to improve its computational accuracy.

(4) From the following options (A–D), choose the one that is NOT an appropriate application of multimodal AI.

A Showing an image and having it described in text.

B Listening to audio and transcribing it into text.

C Giving instructions in text and generating an image.

D Computing the result of input numerical values using only the rules of the four arithmetic operations and returning it.

⚙ Think as an Engineer — Investigate & Decide

A school club wants to use a generative-AI chatbot to help write short articles about local history for the school website. You advise on using it responsibly.

☐ **Publish whatever the AI writes** ☐ **Use the AI, but verify before publishing**

1 · Decide where it helps. Name one part of the task where the chatbot genuinely saves effort, and one part where its output must not be trusted as-is. Explain each in one line.

2 · Analyze the risk. Using how a language model chooses words, explain why the chatbot might state a wrong date or invented name for a local landmark very confidently, and describe exactly how the club would catch such a hallucination before publishing — a check the lesson does not spell out for you.

3 · Decide the safeguards. State the rule the club should follow for every AI-written article before it goes live, and say what role RLHF-style tuning does and does not solve here. **In pairs,** compare rules and agree on the one check you would never skip.

Give evidence for your answer.

KEY TERMS

generative AI — AI that creates new text, images, or audio.

fact-check — verifying a claim against a reliable source.

Stuck? Remember the model is rewarded for *sounding* right, not for *being* right. Which items in a history article — dates, names, places — are exactly the kind of thing it could get confidently wrong? Think about who or what could confirm each one, and at what point a person should step in.

🌎 In a New Context

A student uploads a photo of a handwritten note and asks an AI assistant to read it aloud and then summarize it in text. Name the property that lets one model move between image, text, and audio, and explain in one line how the network can handle all three.

❓ Lesson Question — Answered

The lesson asked how a language model generates text, what makes a model "large," and why it can still get facts wrong. A language model learns from large amounts of text to predict the likely next word, and it generates text by repeating that prediction, connecting one word at a time into a whole sentence. It chooses each word by a probability of coming next — the statistically plausible word, not the guaranteed-correct one — which is exactly why hallucination (fluent but false responses) can occur. A large language model scales this up: trained on vast Internet text with billions to trillions of parameters, and beyond a certain scale it gains qualitatively new abilities such as reasoning, summarizing, translating, and generating code (ChatGPT, Claude, and Gemini are examples). Because raw training does not guarantee helpful answers, RLHF tunes the model with human evaluations as rewards, making responses natural, on-intent, and safer. And multimodality lets a single model handle text, images, and audio together by converting them all into a common numerical representation. The lasting caution: fluency is not truth, so outputs must still be checked.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A model that predicts which word is likely to come next in a sentence is called a (a). Text generation is achieved by repeating the prediction of the (b). A language model based on a very large neural network trained using vast

⭐ REFLECT & CHALLENGE

Reflect: When you last used a generative-AI tool, did you check its facts? What could a confident wrong answer have cost? **Challenge:** A classmate says "RLHF makes the AI always tell the truth." Explain why that is not correct, and state what RLHF actually improves.

amounts of text from the Internet is called a (c); when the number of (d) or the amount of training data exceeds a certain scale, qualitatively new abilities sometimes appear. Since just learning from vast text does not necessarily lead to responses desirable for humans, (e), a method of tuning by reinforcement learning using human evaluations as rewards, is used. Also, the property of being able to handle multiple kinds of data—text, images, audio, video, and so on—with a single model is called (f).

1. Fill in the blanks (a)–(f).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the mechanism of text generation.

A Text is generated by connecting the results of repeatedly predicting the next word.

B Text is generated by selecting from thousands of prepared templates.

C Text is generated by first deciding the length of the sentence and then deciding all words simultaneously.

D Text is generated by machine-translating the input sentence.

(2) From the following options (A–D), choose the one that most appropriately describes the background to large language models acquiring new abilities.

A Because reducing the number of parameters below a certain level simplified the processing.

B Because limiting training data to a small amount of high-quality text raised quality.

C Because qualitatively new abilities appear when the number of parameters or the amount of training data exceeds a certain scale.

D Because humans wrote down the rules of language one by one and taught them to the model.

(3) From the following options (A–D), choose the one that most appropriately describes the purpose for which RLHF is performed.

A To increase the number of words a language model can handle.

B To make a language model's responses natural and aligned with the intent of the question.

C To enable a language model to handle data other than text.

D To reduce the number of parameters used in a language model's calculations.

⭐ Key takeaway

Remember:
A language model generates text by predicting the next word (plausible, not always true → hallucination). LLMs use vast data and billions to trillions of parameters; RLHF tunes them by human feedback; multimodality unifies text, image, and audio. Fluency ≠ truth — always verify.